

1.) A 1-m long dipole is excited by a 1 MHz current with an amplitude of 12 A. What is the average power density radiated by the dipole at a distance of 5 km in a direction that is 45° from the dipole axis? (8 points)

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{1 \times 10^6 \text{ Hz}} = 300 \text{ m}$$

$$\frac{l}{\lambda} = \frac{1}{300} = \frac{1}{300}, \text{ which is } \ll \frac{1}{50} \text{ so Hertzian}$$

Avg power density

$$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32 \pi^2 R^2} \sin^2(\theta)$$

$$= \frac{120 \pi \left(\frac{\pi}{150}\right)^2 (12)^2 (1)^2}{32 \pi^2 5000^2} \sin^2(45^\circ)$$

nW/m²

$$S(5000, 45^\circ) = 1.51 \text{ pW/m}^2$$

Given: $\eta_0 = 120 \pi$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{300} = \frac{\pi}{150}$$

$$\lambda = 300$$

$$I_0 = 12 \text{ A}$$

$$l = 1 \text{ m}$$

$$R = 5000 \text{ m}$$

$$\theta = 45^\circ = \frac{\pi}{4} \text{ rads}$$

2.) The normalized radiation intensity of a certain antenna is given by

$$F(\theta) = \exp(-20\theta^2) \text{ for } 0 \leq \theta \leq \pi,$$

where θ is in radians. Determine: (15 points total)

a. the half power beamwidth, (5 points) (solve for β) (rads mode)

$$\begin{aligned} F(\theta_{1/2}) &= 0.5 \\ e^{-20\theta_{1/2}^2} &= 0.5 \\ -20\theta_{1/2}^2 &= \ln(0.5) \end{aligned}$$

$$\theta_{1/2} = \pm 0.1862 \text{ rads}$$

$$\beta = 2\theta_{1/2} = 2(0.1862) = \boxed{0.3723 \text{ rads}}$$

b. the pattern solid angle, (5 points) Ω_p

$$\begin{aligned} \Omega_p &= \int_0^{2\pi} \int_0^\pi F(\theta) \sin(\theta) d\theta d\phi \\ &= 2\pi \int_0^\pi e^{-20\theta^2} \sin(\theta) d\theta \end{aligned}$$

use TI-36x Pro

Literally plug it in as is

but use X instead of θ

$$\boxed{\Omega_p = 0.156 \text{ sr}}$$

$$(155.7771574 \times 10^{-3})$$

c. the antenna directivity. (5 points)

$$D = \frac{4\pi}{\Omega_p} = \frac{4\pi}{0.156} = \boxed{80.67 \text{ (unitless)}}$$

- 3.) For a dipole antenna of length $l = \lambda/4$, (15 points total)
 a. determine the directions of maximum radiation, (5 points)

$$\lim_{\theta \rightarrow 0} \frac{f(\theta)}{g(\theta)} = \lim_{\theta \rightarrow 0} \frac{f'(\theta)}{g'(\theta)}$$

Base formula

$$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$$

$$l = \frac{\lambda}{4}$$

$$\frac{\pi l}{\lambda} = \frac{\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{4}$$

$$\frac{\pi l}{\lambda} = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \cos(\frac{\pi}{4})}{\sin(\theta)} \right]^2$$

$$= \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin(\theta)} \right]^2$$

$$(S(\theta) \text{ is directly proportional to } \frac{15 I_0^2}{\pi R^2})$$

constant w.r.t θ so let $= S_0$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin(\theta)} \right]^2$$

Use ti-36x pro

$$\left. \frac{d}{dx} \left(\cos\left(\frac{\pi}{4} \cos(x)\right) - \frac{\sqrt{2}}{2} \right) \right|_{x=x} = 0$$

$$\left. \frac{d}{dx} (\sin \theta) \right|_{x=x} = 999.9 \times 10^{-3} = 1$$

$$S(\theta) = S_0 \left[\frac{0}{1} \right]^2$$

$$S(\theta) = 0 \text{ (peak is somewhere between } 0, \pi)$$

use calc

θ	30°	60°	90°
S/S_0	0.020	0.063	0.0860

$$\theta = 90^\circ = \frac{\pi}{2}$$

- b. obtain an expression for $S_{\max}$, (5 points)

$$S_{\max} = S(\theta)$$

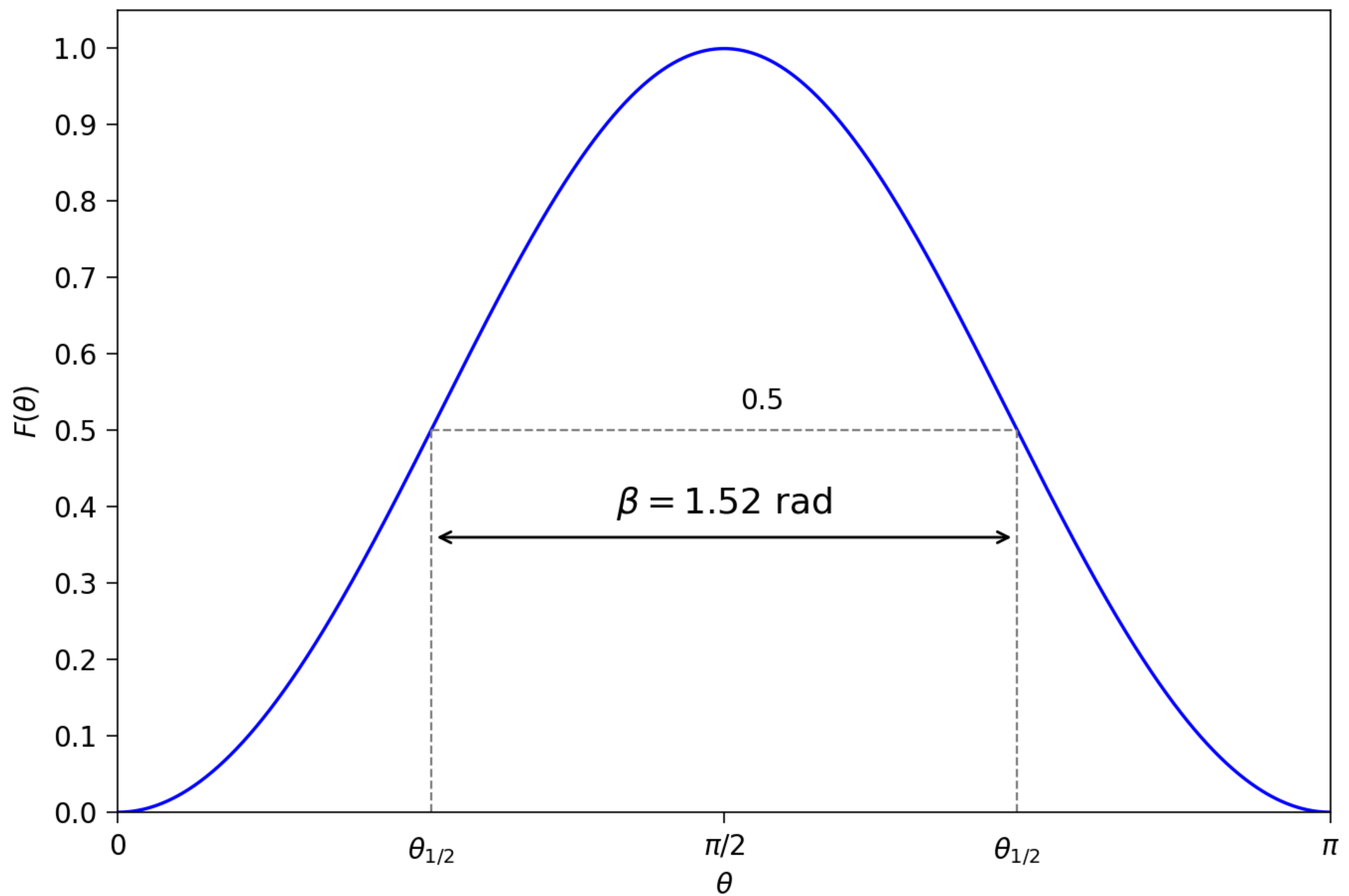
$$S\left(\frac{\pi}{2}\right) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos\left(\frac{\pi}{2}\right)\right) - \frac{\sqrt{2}}{2}}{\sin\left(\frac{\pi}{2}\right)} \right]^2$$

$$= 0.0858 \frac{15 I_0^2}{\pi R^2}$$

c. generate a plot of the normalized radiation pattern $F(\theta)$. (5 points)

$$F(\theta) = \frac{S(\theta)}{S_{\max}} = \frac{\frac{15 I_0^2}{\pi R^2} \left[\frac{\cos\left(\frac{\pi}{4} \cos(\theta)\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)} \right]^2}{\frac{15 I_0^2}{\pi R^2} 0.0858}$$

$$= 11.66 \left[\frac{\cos\left(\frac{\pi}{4} \cos(\theta)\right) - \frac{\sqrt{2}}{2}}{\sin(\theta)} \right]^2$$



4.) Determine the effective area of a half-wave dipole antenna at 100 MHz, and compare it with its physical cross section if the wire diameter is 2 cm. (5 points)

$$\text{Wave length } \lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{100 \times 10^6 \text{ Hz}} = 3 \text{ m}$$

Since half wave dipole $D = 1.64$ from cheat sheet

Effective Area

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ m}^2$$
$$= \frac{(3^2)(1.64)}{4\pi}$$

$$A_e = 1.17 \text{ m}^2$$

Physical cross section

} for half wave dipole is $l = \frac{\lambda}{2}$, $d = 0.02 \text{ m}$

$$A_p = l \cdot d = \frac{\lambda}{2} d = \frac{3}{2} (0.02) = 0.03 \text{ m}^2$$

Ratio

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = 39.2$$

The effective area is $39 \times$ larger than physical cross section.

5.) A half-wave dipole TV broadcast antenna transmits 1 kW at 50 MHz. What is the power received by a home television antenna with a 3 dB gain if located at a distance of 30 km? (5 points)

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{50 \times 10^6} = 6 \text{ m}$$

Gain transmitting

$G_t = D$ for lossless so from chart sheet for half wave dipole

$$D = 1.64 = G_t$$

Gain receiving

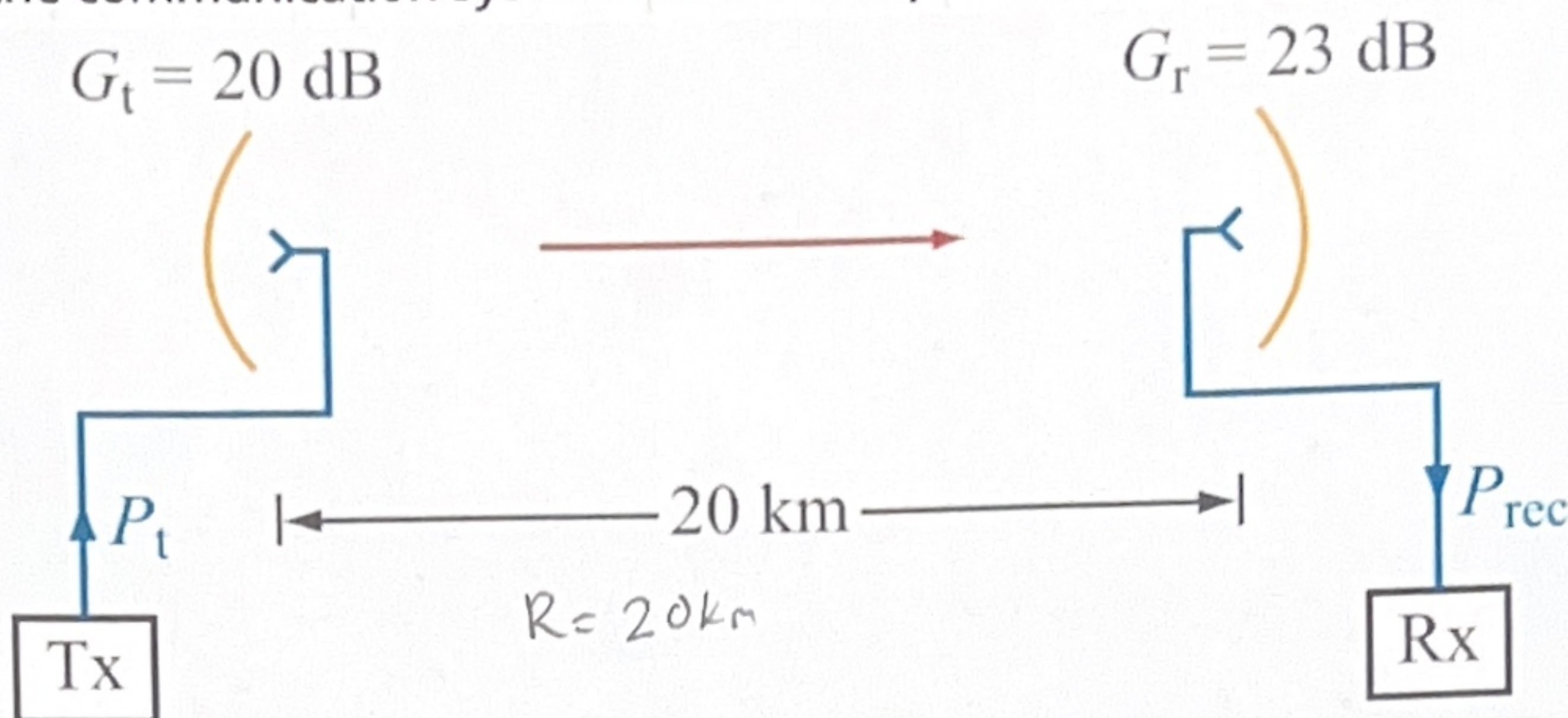
$G_r = 3 \text{ dB (given)} \rightarrow \text{convert to linear} = G_{r \text{ linear}} = 10^{3 \text{ dB}/10} \approx 2$

$$R = 30 \text{ km} = 30 \times 10^3 \text{ m}$$

$$P_t = 1 \text{ kW} = 1000 \text{ W}$$

$$P_r = (1.64) (2) \left(\frac{6}{4\pi (30 \times 10^3)} \right)^2 (1000) = \boxed{8.31 \text{ nW}}$$

6. Consider the communication system with all components properly matched shown below.



Determine the following if $P_t = 10$ W and $f = 6$ GHz.

(a) What is the power density at the receiving antenna (assuming proper alignment of antennas)?

$$S_r = \frac{G_t P_t}{4\pi R^2} = \frac{(100)(10)}{4\pi(20 \times 10^3)^2} = 198.9 \text{ nW/m}^2$$

$G_{t, \text{linear}} = 10^{20/10} = 100$

(b) What is the received power?

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$$

$$= (100)(199.5) \left(\frac{0.05}{4\pi(20 \times 10^3)} \right)^2 10$$

$$= 7.8969 \times 10^{-9}$$

$$P_r = 7.90 \text{ nW}$$

$$G_r = 10^{23/10} = 199.5$$

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{6 \times 10^9} = 50 \times 10^{-3} \text{ m} = 0.05$$

(c) If $T_{\text{sys}} = 1000$ K and the receiver bandwidth is 20 MHz, what is the signal-to-noise ratio in decibels?

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{P_r}{P_N}$$

$$= 10 \log_{10} \frac{7.90 \text{ nW}}{276 \times 10^{-15}}$$

$$= 44.6 \text{ dB}$$

$$P_N = k_B T_{\text{sys}} B$$

$$= (1.38 \times 10^{-23})(1000)(20 \times 10^6)$$

$$= 276 \times 10^{-15} \text{ W}$$

7. An antenna with a circular aperture has a circular beam with a beamwidth of 3° at 20 GHz.

(a) What is the antenna directivity in dB?

$$\beta = 3^\circ \cdot \frac{\pi}{180} = 0.05236 \text{ rad}$$

$$D_{\text{lin}} = \frac{4\pi}{\beta^2} = \frac{4\pi}{(0.05236)^2} = 4584 \text{ unitless}$$

$$D_{\text{dB}} = 10 \log(4584) = \boxed{36.6 \text{ dB}}$$

(b) If the antenna area is doubled, what will be the new directivity and new beamwidth?

$$D_{\text{new}} = 2D_{\text{old}} = 2(4584) = 9168 \text{ unitless}$$

$$D_{\text{new dB}} = 10 \log_{10}(9168) = \boxed{39.6 \text{ dB}}$$

$$\beta_{\text{new}} = \frac{\beta_{\text{old}}}{\sqrt{2}} = \frac{0.05236 \text{ rad}}{\sqrt{2}} = 0.037 \text{ rad}$$

$$\beta_{\text{new}}^\circ = 0.037 \times \frac{180}{\pi} = \boxed{2.12^\circ}$$

(c) If the aperture is kept the same as in (a), but the frequency is doubled to 40 GHz, what will the directivity and beamwidth become then?

$$f_{\text{new}} = 40 \times 10^9 \text{ Hz}$$

$$D_{\text{new}} = 4D_{\text{old}} = 4(4584) = 18336$$

$$D_{\text{new dB}} = 10 \log(D_{\text{new}}) = \boxed{42.63 \text{ dB}}$$

$$\beta_{\text{new}}^\circ = \frac{\beta_{\text{old}}}{2} = \frac{3^\circ}{2} = \boxed{1.5^\circ}$$

8. Find and plot the normalized array factor and determine the half-power beamwidth for a 5-element linear array excited with equal phase and a uniform amplitude distribution. The interelement spacing is $3\lambda/4$.

$$N = 5$$

$$k = \frac{2\pi}{\lambda}$$

$$d = \frac{3\lambda}{4}$$

$$\gamma = kd \cos \theta$$

$$d = \frac{3\lambda}{4}$$

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$$

$$\gamma = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} \cos(\theta) = \frac{3\pi}{2} \cos \theta$$

$$= \frac{\sin^2\left(\frac{5 \left(\frac{3\pi}{2}\right) \cos \theta}{2}\right)}{\sin^2\left(\frac{\frac{3\pi}{2} \cos \theta}{2}\right)}$$

$$= \frac{\sin^2\left(\frac{15\pi}{4} \cos \theta\right)}{\sin^2\left(\frac{3\pi}{4} \cos \theta\right)}$$

$$F_{\text{norm}} = \frac{F_a}{N^2} =$$

$$N^2 = 5^2 = 25$$

$$F_{\text{norm}} = \frac{1}{25} \frac{\sin^2\left(\frac{15\pi}{4} \cos \theta\right)}{\sin^2\left(\frac{3\pi}{4} \cos \theta\right)}$$

Uniform Broadside array

$$\beta = 0.886 \frac{\lambda}{Nd} \text{ (rad)}$$

$$Nd = 5 \cdot \frac{3\lambda}{4} = \frac{15\lambda}{4}$$

$$\beta = 0.886 \cdot \frac{\lambda}{\frac{15\lambda}{4}}$$

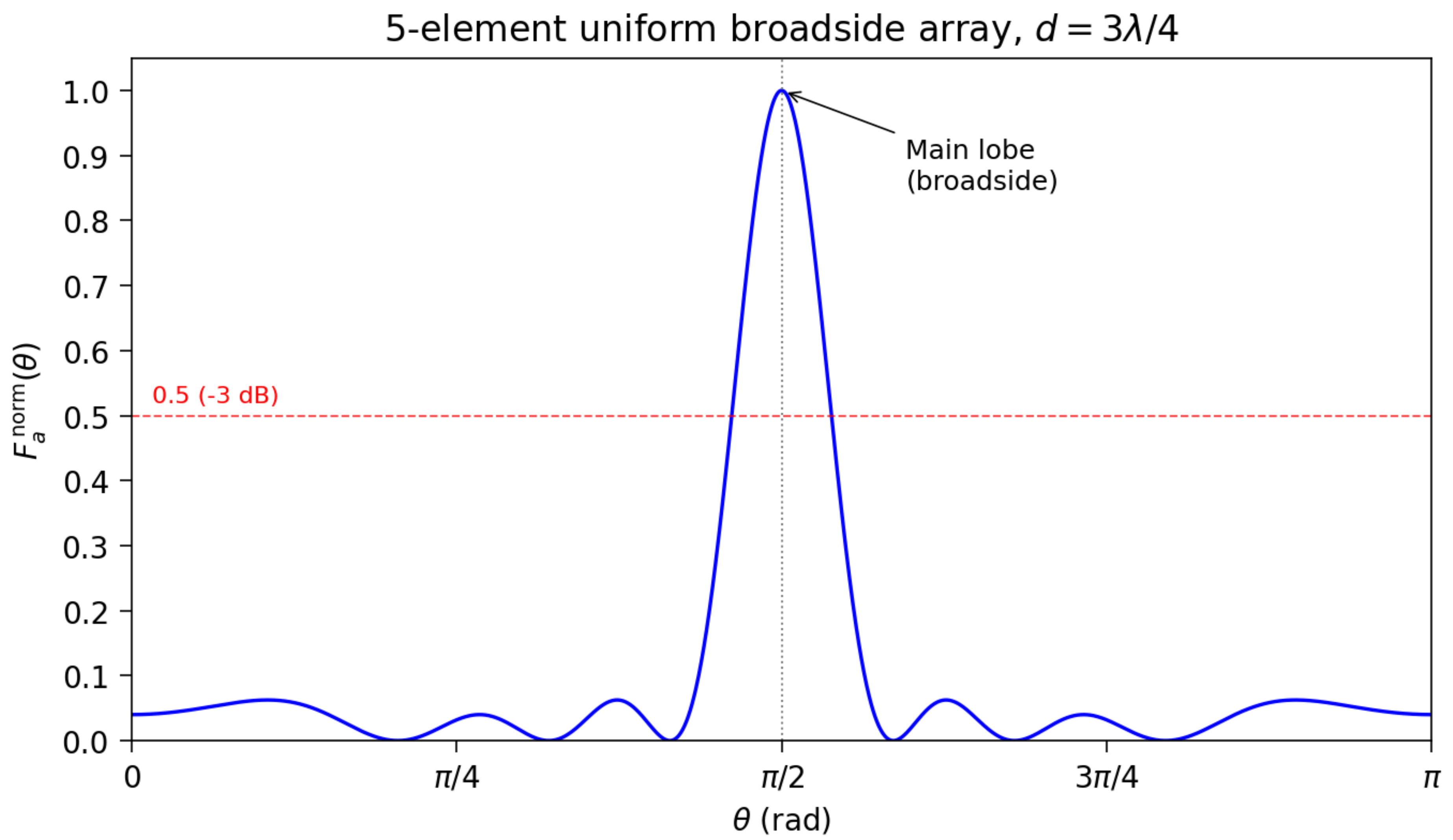
$$= 0.886 \cdot \cancel{\lambda} \cdot \frac{4}{15\cancel{\lambda}}$$

$$= 0.2363 \text{ rad}$$

$$\beta_{\text{deg}} = \beta \left(\frac{180}{\pi}\right)$$

$$= 0.2363 \left(\frac{180}{\pi}\right)$$

$$= \boxed{13.5^\circ}$$



HPBW (from plot) ~ 13.5 deg (matches $\beta = 0.886 \lambda / (Nd) = 0.236$ rad calculation above on page 9).
Main lobe at $\theta = \pi/2$ (broadside, peak = 1). Side lobes < 0.06 .